



University of Dhaka

Department of Biomedical Physics and Technology

BMPT 514

Biomedical Instrumentation Laboratory

Experiment No. 02

Study the Different Stages of an EMG Acquisition Circuit and the Measurement of EMG

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Experiment No. 02

2.1 Name of the Experiment

Study the Different Stages of an EMG Acquisition Circuit and the Measurement of EMG

2.2 Objectives

- To get acquainted with the circuit diagram and data acquisition process in the EMG trainer kit.
- To observe the trace of EMG.

2.3 Theory

Electromyography (EMG) is the technique used to record and visualize the electrical activity generated by skeletal muscles. A muscle consists of numerous muscle fibers. In the resting state, each muscle fiber maintains a polarized membrane potential due to charge separation across its membrane. When stimulated by a motor neuron, the membrane depolarizes and generates an action potential that propagates along the muscle fiber.

The combined electrical activities of multiple muscle fibers create electrical signals known as EMG signals. These signals are generated when motor units are activated during voluntary muscle contraction. Since the human body behaves as a volume conductor, the electrical activity produced by muscles can be detected at the skin surface using electrodes.

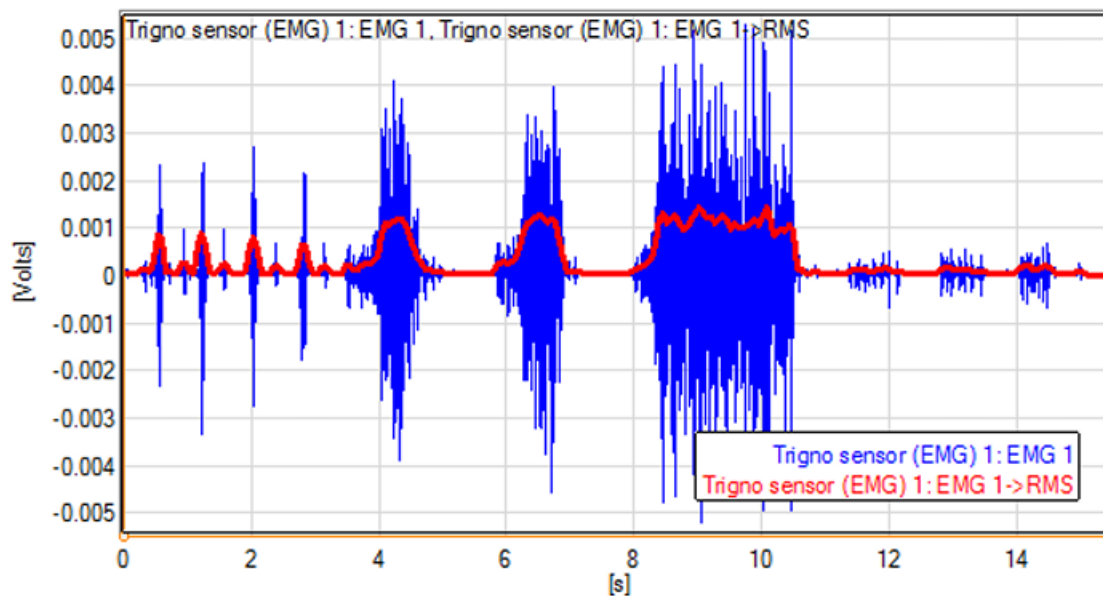


Fig. 01: Typical EMG signal for voluntary muscle contraction

Module Diagram

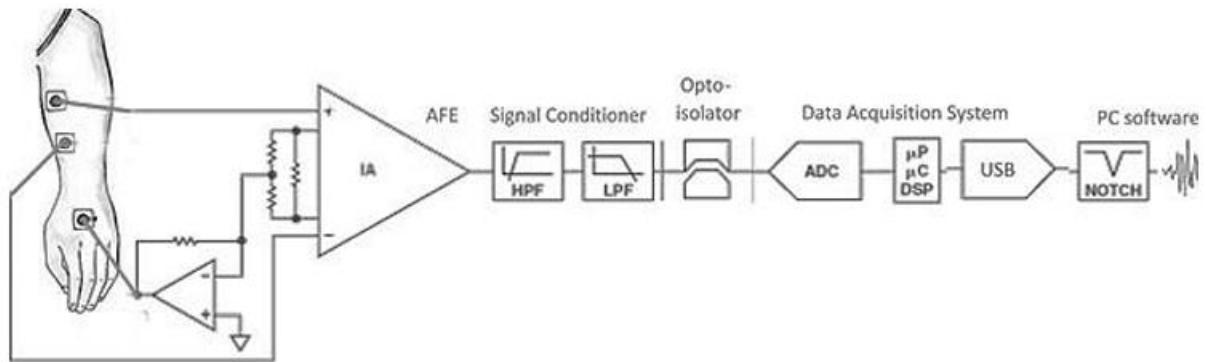


Fig. 02: Block diagram for EMG trainer module

2.4 Equipment

- EMG trainer module
- Computer
- Multimeter

2.5 Setup

Practical set-up is shown in the following figure.

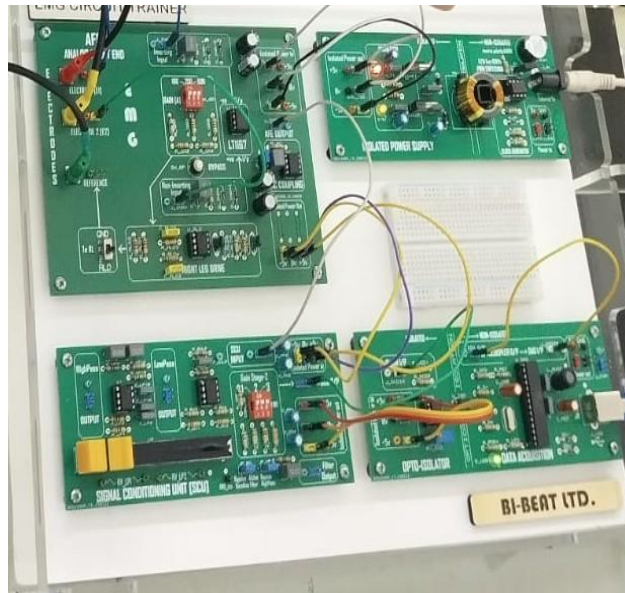
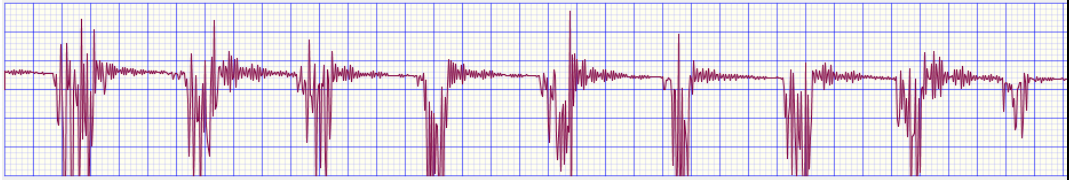


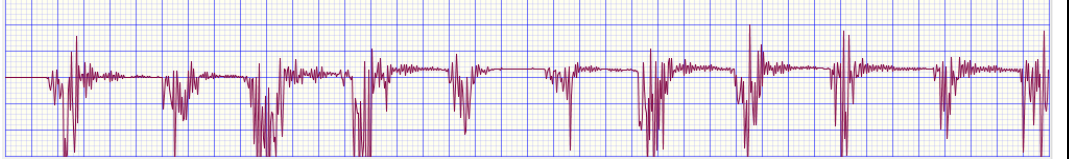
Fig. 03: Practical set-up for EMG acquisition

2.6 EMG Plot

Specimen 1

Activity	EMG plot
Index finger movement	

Specimen 2

Activity	EMG plot
Thumb finger movement	

2.7 Discussion

In this experiment, the EMG acquisition circuit was studied by recording muscle activity during voluntary finger movement. The EMG trainer module was able to detect the small electrical signals produced by skeletal muscles and display them as visible waveforms. During the resting condition, the signal level was comparatively low, while clear bursts appeared during finger movement. This shows that muscle contraction produces measurable electrical activity at the skin surface.

The recorded EMG plots for index finger and thumb finger movement showed repeated peaks with different amplitudes. The index finger movement produced stronger and more frequent signal changes in some portions, while the thumb movement also showed clear muscle activation but with a slightly different waveform pattern. This difference is expected because different fingers involve different muscle groups and different levels of contraction force.

Some irregularities and noise were observed in the EMG traces, which may have occurred due to electrode placement, hand movement, skin-electrode contact, or external electrical interference. Since EMG signals are very weak, even small movements or poor contact can affect the recorded waveform. Overall, the experiment successfully demonstrated the basic operation of an EMG acquisition system and helped to understand how voluntary muscle activity can be recorded and observed using surface electrodes.